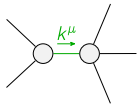


|                                                         |                                      |                                               |                                           |                                                         |                                           |                                               |
|---------------------------------------------------------|--------------------------------------|-----------------------------------------------|-------------------------------------------|---------------------------------------------------------|-------------------------------------------|-----------------------------------------------|
|                                                         | $\mathcal{K}_{0^+}^{\#1}$            | $\mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi}$ |                                           |                                                         |                                           |                                               |
| $\mathcal{K}_{0^+}^{\#1}{}^\dagger$                     | 0                                    | 0                                             |                                           |                                                         |                                           |                                               |
| $\mathcal{K}_{0^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$ | 0                                    | 0                                             | $\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$               | $\mathcal{K}_{1^+}^{\#1}{}_{\alpha}$      | $\mathcal{K}_{1^+}^{\#2}{}_{\alpha}$          |
| $\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta}$     | $-k^2 \frac{(4)}{\kappa_3}$          | $-\frac{k^2 (4)}{\sqrt{2} \kappa_3}$          | 0                                         | 0                                                       |                                           |                                               |
| $\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta}$     | $-\frac{k^2 (4)}{\sqrt{2} \kappa_3}$ | $-\frac{k^2 (4)}{2 \kappa_3}$                 | 0                                         | 0                                                       |                                           |                                               |
| $\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha}$          | 0                                    | 0                                             | 0                                         | 0                                                       |                                           |                                               |
| $\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha}$          | 0                                    | 0                                             | 0                                         | 0                                                       | $\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi}$ |
|                                                         |                                      |                                               |                                           | $\mathcal{K}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta}$     | 0                                         | 0                                             |
|                                                         |                                      |                                               |                                           | $\mathcal{K}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$ | 0                                         | 0                                             |

|                                                         |                                          |                                               |                                           |                                                         |                                           |                                               |
|---------------------------------------------------------|------------------------------------------|-----------------------------------------------|-------------------------------------------|---------------------------------------------------------|-------------------------------------------|-----------------------------------------------|
|                                                         | $\mathcal{J}_{0^+}^{\#1}$                | $\mathcal{J}_{0^+}^{\#1}{}_{\alpha\beta\chi}$ |                                           |                                                         |                                           |                                               |
| $\mathcal{J}_{0^+}^{\#1}{}^\dagger$                     | 0                                        | 0                                             |                                           |                                                         |                                           |                                               |
| $\mathcal{J}_{0^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$ | 0                                        | 0                                             | $\mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta}$               | $\mathcal{J}_{1^+}^{\#1}{}_{\alpha}$      | $\mathcal{J}_{1^+}^{\#2}{}_{\alpha}$          |
| $\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta}$     | $-\frac{4}{9 k^2 (4) \kappa_3}$          | $-\frac{2 \sqrt{2}}{9 k^2 (4) \kappa_3}$      | 0                                         | 0                                                       |                                           |                                               |
| $\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta}$     | $-\frac{2 \sqrt{2}}{9 k^2 (4) \kappa_3}$ | $-\frac{2}{9 k^2 (4) \kappa_3}$               | 0                                         | 0                                                       |                                           |                                               |
| $\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha}$          | 0                                        | 0                                             | 0                                         | 0                                                       |                                           |                                               |
| $\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha}$          | 0                                        | 0                                             | 0                                         | 0                                                       | $\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta\chi}$ |
|                                                         |                                          |                                               |                                           | $\mathcal{J}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta}$     | 0                                         | 0                                             |
|                                                         |                                          |                                               |                                           | $\mathcal{J}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$ | 0                                         | 0                                             |

### Lagrangian

$${}^{(4)}\kappa_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^\delta - 2 {}^{(4)}\kappa_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta - \kappa_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta - \frac{1}{2} {}^{(4)}\kappa_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta$$

| Added source term(s):                                                              | $\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                               |
|------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| Source constraint(s)                                                               | # constraint(s)                                               | Covariant form                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                               |
| $\mathcal{J}_{0^+}^{\#1\alpha\beta\chi} == 0$                                      | 1                                                             | $\partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} ==$<br>$\partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                               |
| $\mathcal{J}_{0^+}^{\#1} == 0$                                                     | 1                                                             | $\partial_\beta \mathcal{J}^\alpha{}_\alpha{}^\beta == 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                               |
| $\mathcal{J}_{1^+}^{\#2\alpha} == 0$                                               | 3                                                             | $\partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi} == 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                               |
| $\mathcal{J}_{1^+}^{\#1\alpha} == 0$                                               | 3                                                             | $\partial_\chi \partial^\alpha \mathcal{J}^\beta{}_\beta{}^\chi + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}{}_\beta == \partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                               |
| $\mathcal{J}_{1^+}^{\#1\alpha\beta} == \mathcal{J}_{1^+}^{\#2\alpha\beta}$         | 3                                                             | $3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi} + 2 \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\chi\alpha\beta} == 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                               |
| $\mathcal{J}_{2^+}^{\#1\alpha\beta\chi} == 0$                                      | 5                                                             | $3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta}{}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} +$<br>$4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^\delta{}_\delta{}^\epsilon + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\alpha}{}_\delta ==$<br>$3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha}{}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} +$<br>$2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{J}^\delta{}_\delta{}^\epsilon + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\beta}{}_\delta$ |                               |
| $\mathcal{J}_{2^+}^{\#1\alpha\beta} == 0$                                          | 5                                                             | $3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial^\epsilon \partial_\delta \mathcal{J}^\chi{}_\chi{}^\delta == 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}^\chi{}_\chi{}^\delta + 3 (\partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                               |
| Total # constraint(s):                                                             | 21                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                               |
| Resolved pole(s)                                                                   | # polarization(s)                                             | Square mass                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Residue                       |
|  | 2                                                             | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $-\frac{1}{{}^{(4)}\kappa_3}$ |
| Resolved unitarity condition(s):                                                   |                                                               | ${}^{(4)}\kappa_3 < 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                               |